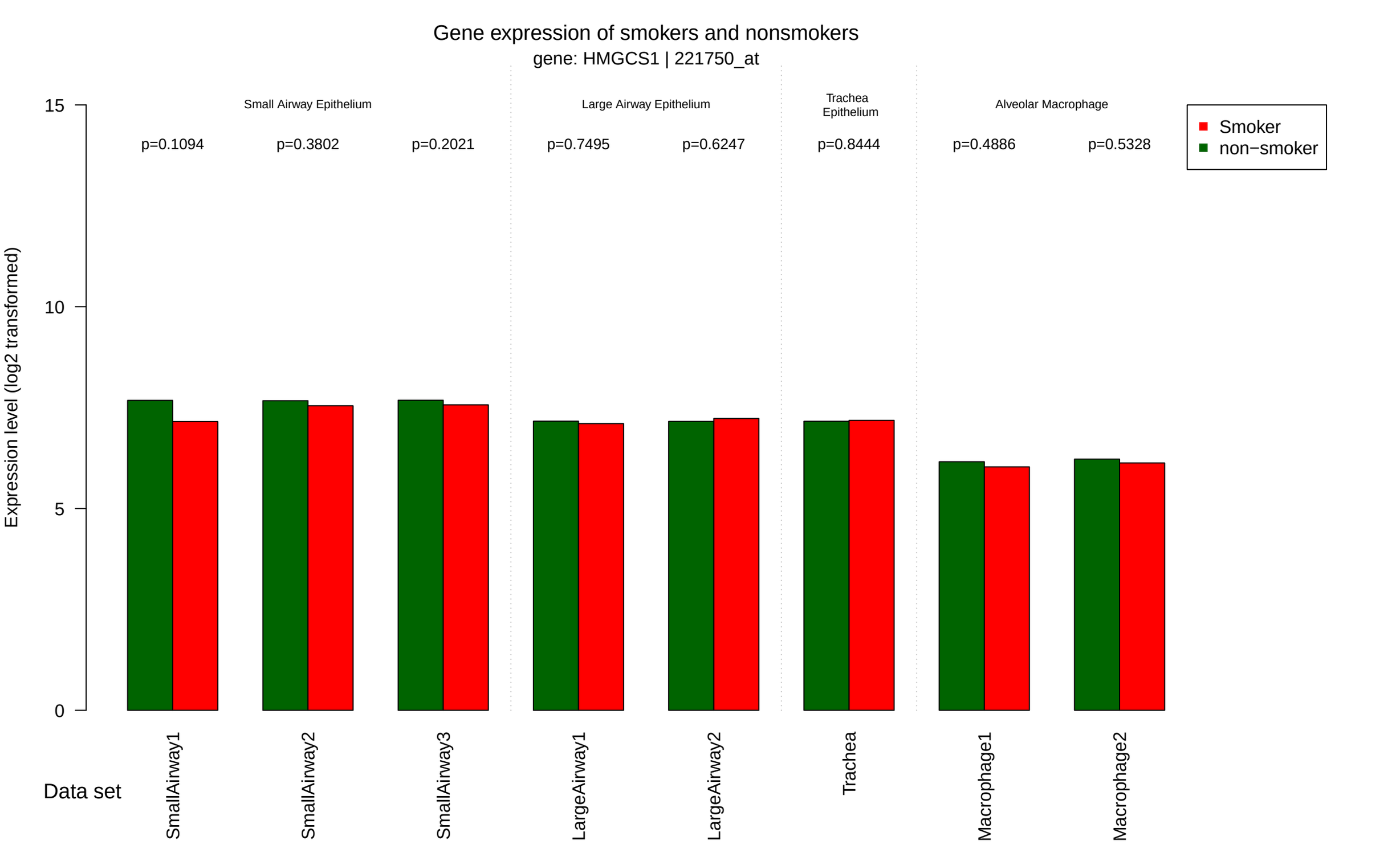
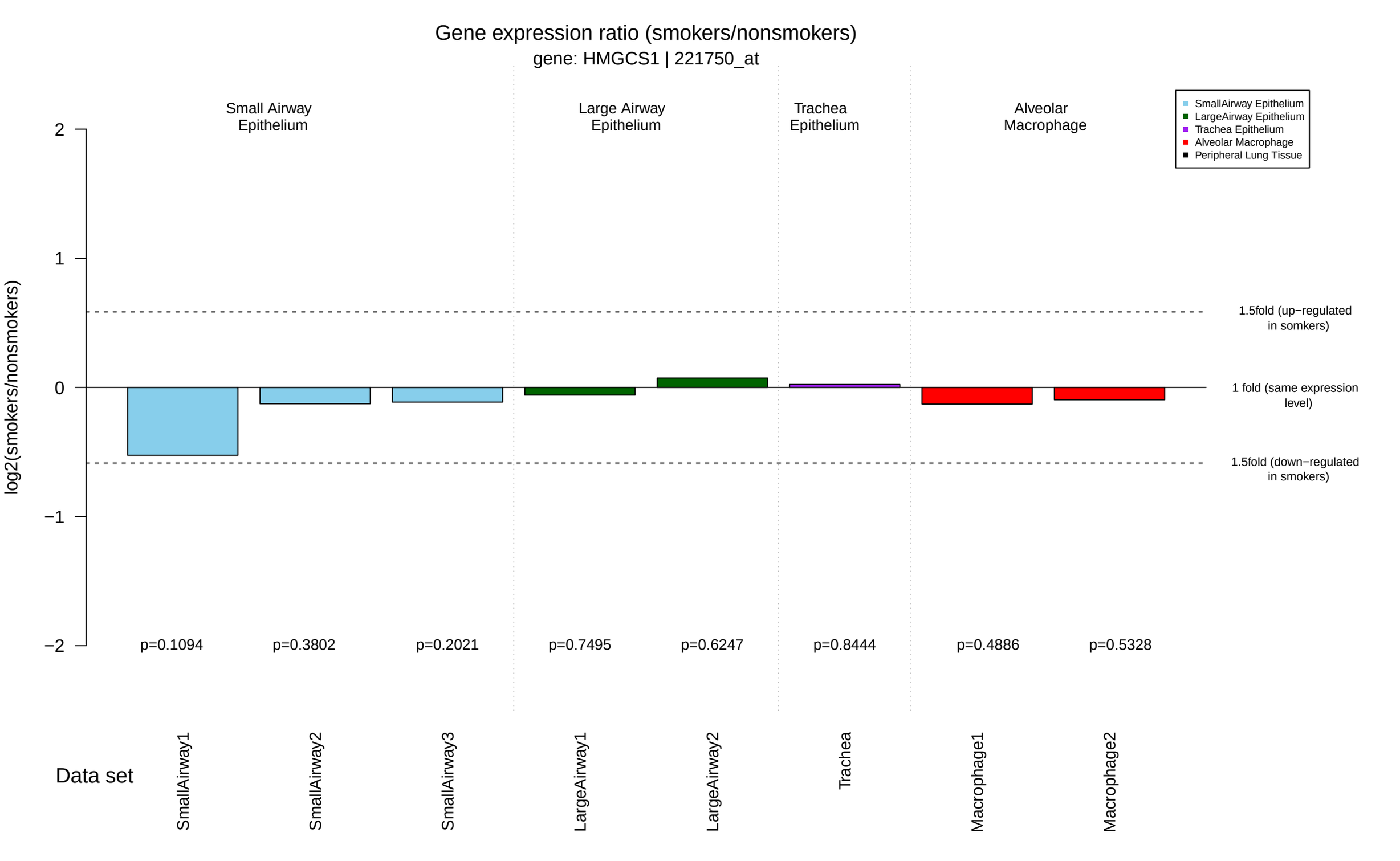
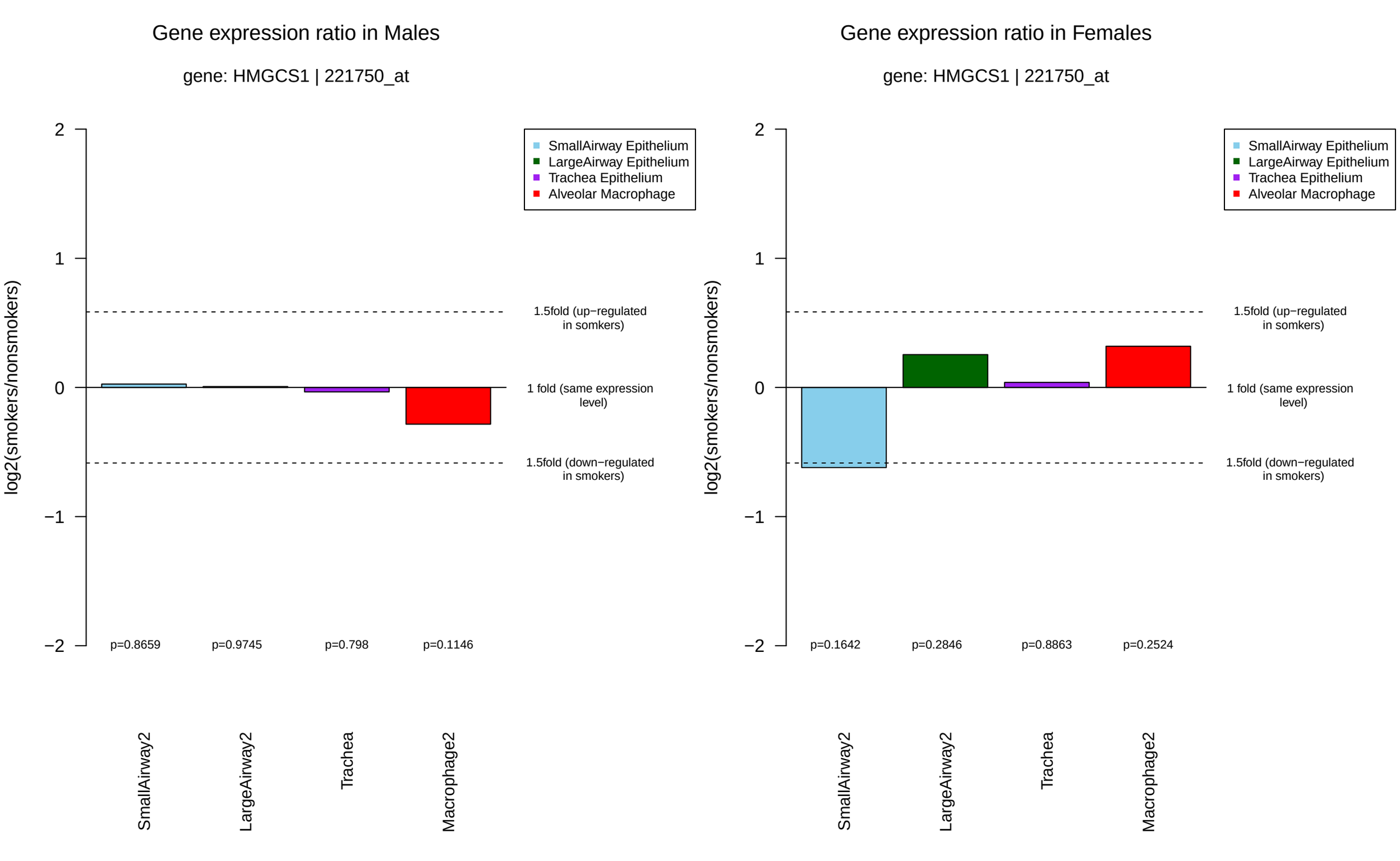
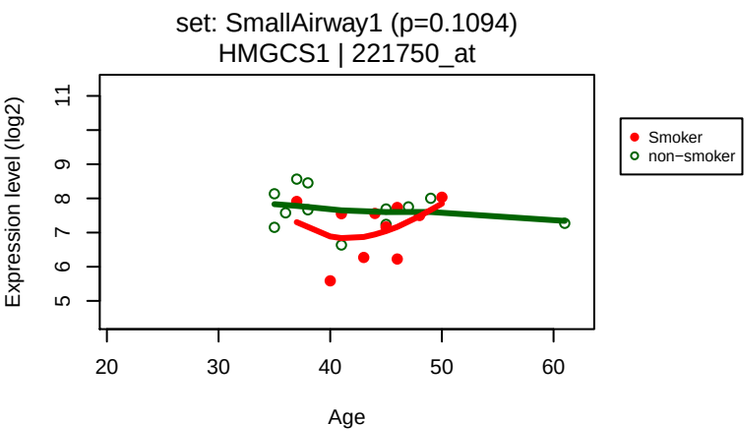
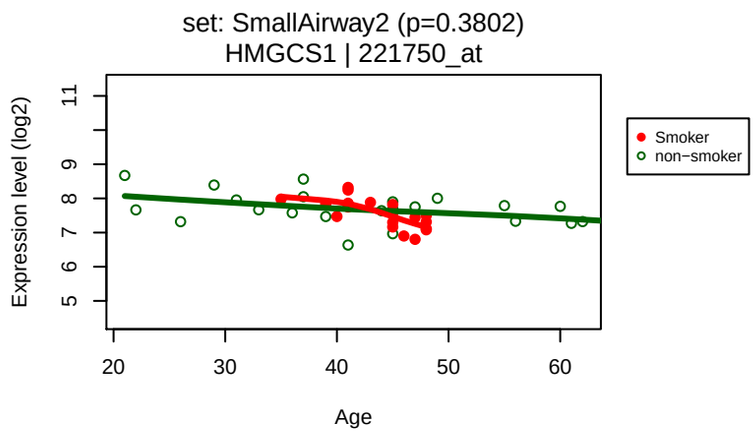
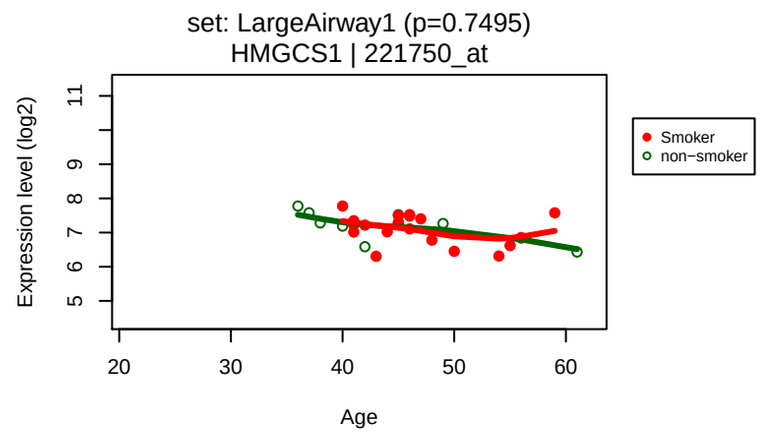

Gene expression of smokers and nonsmokers
gene: HMGCS1 | 221750_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1094
p=0.3802
p=0.2021
p=0.7495
p=0.6247
p=0.8444
p=0.4886
p=0.5328
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

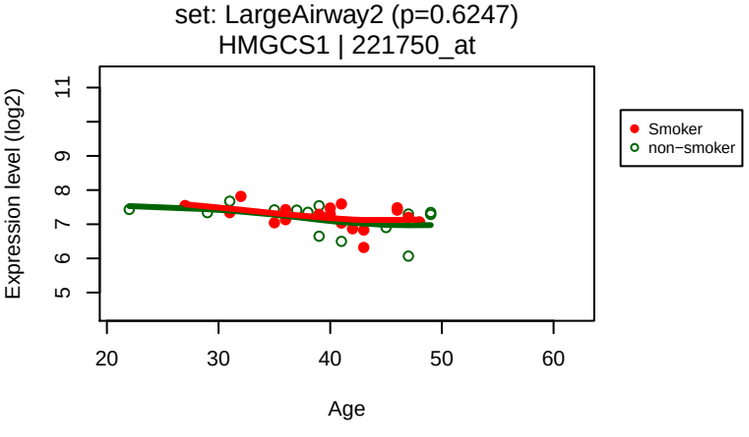

Gene expression ratio (smokers/nonsmokers)
gene: HMGCS1 | 221750_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1094
p=0.3802
p=0.2021
p=0.7495
p=0.6247
p=0.8444
p=0.4886
p=0.5328
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

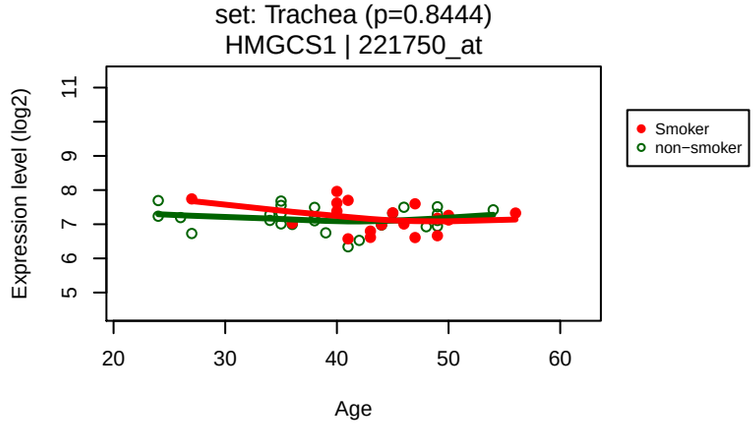

Gene expression ratio in Males
gene: HMGCS1 | 221750_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8659
p=0.9745
p=0.798
p=0.1146
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HMGCS1 | 221750_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1642
p=0.2846
p=0.8863
p=0.2524
SmallAirway2
LargeAirway2
Trachea
Macrophage2

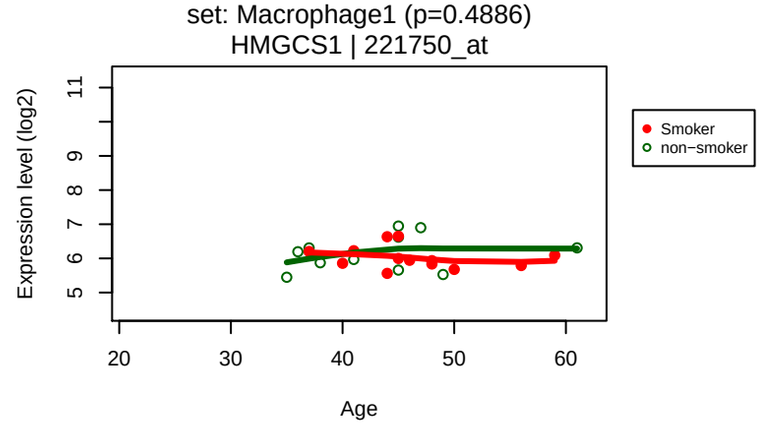

set: SmallAirway1 (p=0.1094)
HMGCS1 | 221750_at
Expression level (log2)
Age
Smoker
non−smoker

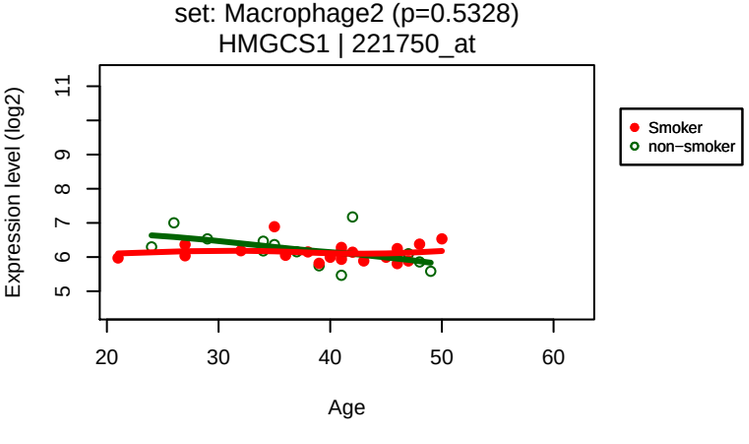

set: SmallAirway2 (p=0.3802)
HMGCS1 | 221750_at
Expression level (log2)
Age
Smoker
non−smoker

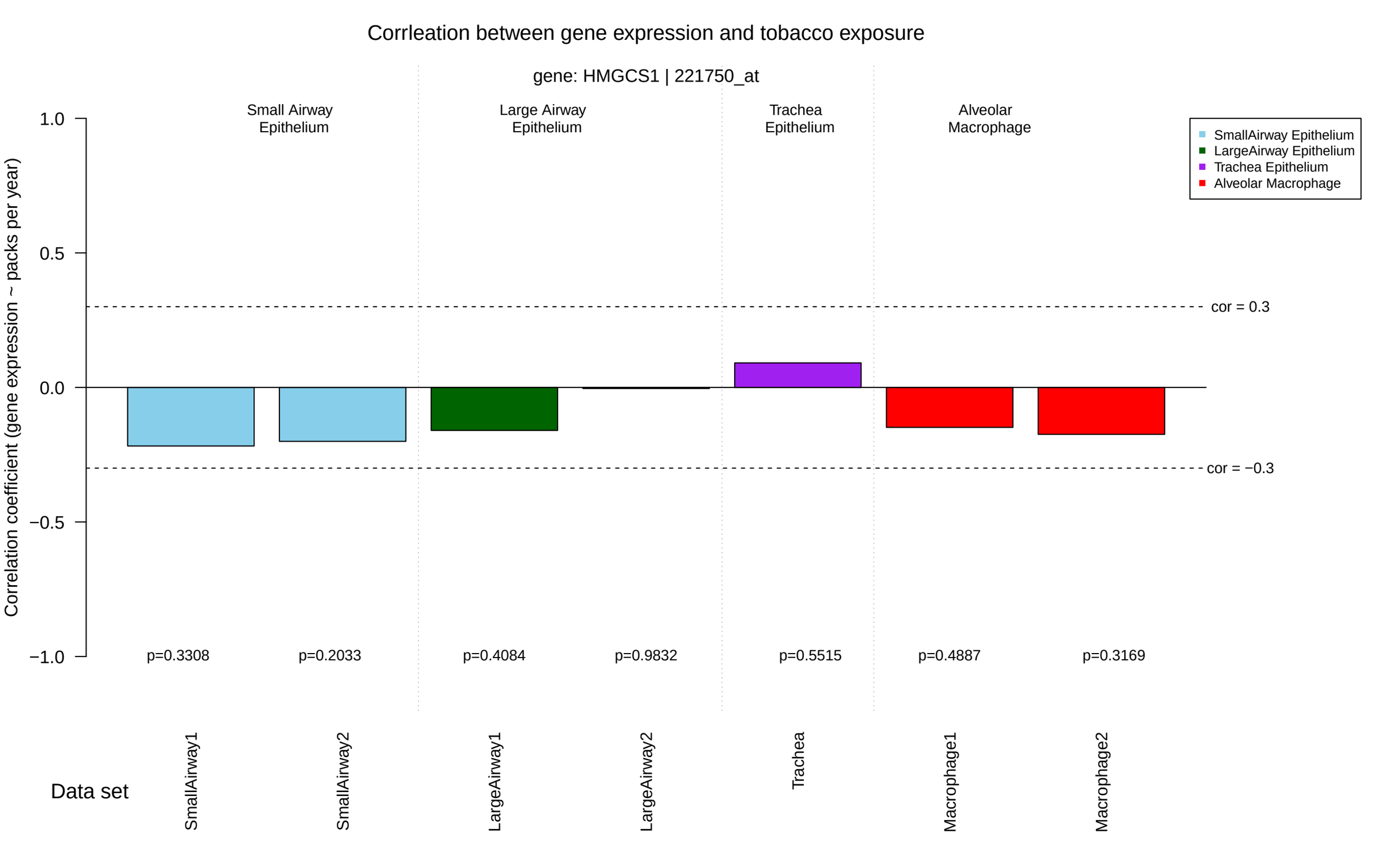

set: LargeAirway1 (p=0.7495)
HMGCS1 | 221750_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6247)
HMGCS1 | 221750_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8444)
HMGCS1 | 221750_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4886)
HMGCS1 | 221750_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5328)
HMGCS1 | 221750_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HMGCS1 | 221750_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3308
p=0.2033
p=0.4084
p=0.9832
p=0.5515
p=0.4887
p=0.3169
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2